Type 3: Software Project

AMRITA VISHWA VIDYAPEETHAM

DEEMED TO BE UNIVERSITY UNDER SECTION 5 OF UGC ACT, 1956

AI-Driven Adaptive Honeypot Platform for Zero-Day Attack Intelligence and Automated Defense

Team Number: 10

S.No	Register Number	Student Name
1	CB.SC.U4CSE23404	Amritha S Nidhi
2	CB.SC.U4CSE23405	Anaswara K
3	CB.SC.U4CSE23429	Madhumithraa T S
4	CB.SC.U4CSE23435	Mayuka Sri K
5	CB.SC.U4CSE23457	Vrithika K M

Guide: **Dr. Harini N**

Associate Professor , Department of CSE

March 26, 2026

Presentation Outline

1. Introduction
2. Risks and Impact
3. Motivation
4. Literature Review
5. Research Gaps
6. Problem Scope, Requirements Feasibility
7. Proposed Solution
8. Novelty of Proposed Solution
9. System Novelty
10. Architecture
11. Timeline of Workplan
12. Implementation - Phase 1
13. References

1. Introduction

- Modern networks face rapidly evolving cyber threats such as ransomware, worms, and advanced persistent attacks.
- **Zero-day attacks** exploit unknown vulnerabilities for which no patch or detection signature exists.
- Signature-based IDS/IPS systems are ineffective during the initial occurrence of such attacks.
- A **honeypot** is a decoy system designed to attract attackers and capture malicious activity.
- Honeypots provide valuable intelligence including login attempts, command sequences, and behavioral patterns.
- There is a need to transform captured intelligence into automated detection and defense mechanisms.

2. Risks and Impact of Zero-Day Attacks

- Rapid propagation of zero-day exploits through automated malware, worms, and botnets, enabling large-scale infection within a short time span.
- Unauthorized data exfiltration, ransomware deployment, privilege escalation, and complete system compromise before defensive mechanisms are activated.
- Significant financial losses due to operational downtime, regulatory penalties, data recovery costs, and long-term reputational damage to organizations.
- Increased incident response time and recovery complexity because detection signatures, patches, and mitigation strategies are unavailable during the initial attack phase.

3. Motivation

- Signature-based IDS/IPS systems primarily rely on predefined attack patterns and known exploit signatures for detection.
- During the first occurrence of a zero-day exploit, these systems lack prior knowledge, resulting in delayed or missed detection.
- Manual signature creation and rule deployment introduce significant response latency, allowing attackers to operate before containment.
- Honeypots provide rich real-world attack intelligence, including login attempts, command sequences, and behavioral patterns.
- There is a need to automatically transform captured attack intelligence into actionable and deployable security rules.
- Objective: Minimize the time gap between attack discovery, analysis, and defensive mitigation through adaptive automation.

4.1. Literature Review – Paper 1

An Adaptable Deep Learning-Based Intrusion Detection System to Zero-Day Attacks

- **Authors:** Mahdi Soltani, Behzad Ousat, Mahdi Jafari Siavoshani, Amir Hossein Jahangir
- **Approach:** Adaptive deep learning–based intrusion detection using open-set recognition.
- **Technique:** Incremental model updates to identify previously unseen zero-day attacks.
- **Contribution:** Improves detection resilience against evolving attack patterns.

Research Gaps:

- Integration of honeypot-derived real attack intelligence to enhance adaptive learning.
- Transformation of learned attack patterns into automatically deployable detection and prevention rules.
- Enterprise-wide sharing of learned intelligence across IDS, IPS, and firewall systems.

4.2. Literature Review – Paper 2

Utilising Deep Learning Techniques for Effective Zero-Day Attack Detection

- **Authors:** Hanan Hindy, Robert Atkinson, Christos Tachtatzis, Jean-Noël Colin, Ethan Bayne, Xavier Bellekens
- **Approach:** Unsupervised deep learning for zero-day attack detection.
- **Technique:** Autoencoder-based anomaly detection without labeled datasets.
- **Contribution:** Effective identification of unknown attacks using network traffic analysis.

Research Gaps:

- Progression toward live attack data collection instead of offline benchmark datasets.
- Inclusion of behavioral and exploit-level intelligence through honeypot interaction.
- Integration of automated response and defense mechanisms beyond anomaly detection.

4.3. Literature Review – Paper 3

Detection of Zero-Day Attacks: An Unsupervised Port-Based Approach

- **Authors:** Agathe Blaise, Mathieu Bouet, Vania Conan, Stefano Secci
- **Approach:** Unsupervised port-centric anomaly detection using distributed monitoring.
- **Technique:** Statistical change-point detection using Modified Z-score on port-level traffic features within a Split-and-Merge Collaborative IDS (CIDS) architecture.
- **Contribution:** Early detection of large-scale zero-day attacks and emerging botnets through distributed port-usage analysis, significantly reducing false positives.

Research Gaps:

- Limited to statistical port-level anomalies (no payload or exploit-level insight).
- No deception-based or honeypot-driven attacker interaction.
- No automated rule generation or real-time defensive deployment.
- Detection-focused, not adaptive or closed-loop..

4.4 Literature Review – Paper 4

AI-enhanced Honeypots for Zero-Day Exploit Detection and Mitigation

- **Author:** Merlin Balamurugan
- **Approach:** AI-integrated honeypot framework for zero-day detection and mitigation.
- **Technique:** Combination of supervised and unsupervised machine learning models trained on datasets such as UNSW-NB15 and CICIDS2017.
- **Contribution:** Achieved 92% detection rate with reduced false positives (3%) and faster response time (2 seconds) compared to traditional honeypots.

Research Gaps:

- Evaluation limited to controlled experimental environments.
- Limited discussion on adversarial robustness of the AI model.
- No automated rule synthesis or closed-loop defense integration.
- Long-term model adaptation and drift handling not addressed.

4.5 Literature Review – Paper 5

A Review of ML-Based Zero-Day Attack Detection

- **Author:** Yang Guo
- **Approach:** Comprehensive survey of Machine Learning-based zero-day attack detection techniques.
- **Technique:** Reviews unsupervised (One-Class SVM, Autoencoders), supervised, hybrid learning, GAN-based, and Transfer Learning approaches.
- **Contribution:** Systematic comparison of ML models, datasets, evaluation metrics, challenges, and future research directions for zero-day detection.

Research Gaps:

- Inconsistent detection accuracy across different attack types.
- Limited and siloed benchmark datasets for fair comparison.
- Lack of real-world validation for true zero-day scenarios.
- Challenges in feature engineering and model generalization.

5. Research Gaps

- Limited integration of honeypot-derived attack intelligence with adaptive machine learning models.
- Lack of automated conversion of learned behavioral patterns into deployable detection and prevention rules.
- Absence of a unified framework for sharing learned intelligence across IDS, IPS, and firewall systems.
- Need for an adaptive feedback mechanism to continuously improve detection capability.

6. Problem Scope, Requirements & Feasibility

Problem Scope

- Focuses on early detection and containment of zero-day attacks using honeypots
- Generates automated detection and prevention rules without patching vulnerabilities

Functional Requirements

- Deploy honeypots simulating common services
- Capture live attack traffic and behavior
- Analyze data using machine learning
- Generate and deploy IDS/IPS rules

Non-Functional Requirements

- Near real-time analysis
- Secure isolation of honeypots from production systems

Feasibility

- Technically feasible using existing security and ML tools
- Phase-wise development fits academic timeline
- Software-based solution with minimal resource requirements

7. Proposed Solution

- Deploy honeypots to capture real-time attacker interactions and malicious activity in a controlled environment.
- Extract behavioral features such as login attempts, command sequences, session duration, and timing characteristics.
- Apply unsupervised machine learning techniques to perform anomaly detection and attack pattern clustering.
- Generate behavioral signatures derived from clustered attack patterns and observed intent.
- Convert generated signatures into IDS and firewall-compatible rules for enforcement.
- Simulate automated defensive deployment to evaluate containment and response effectiveness.
- Enable Social IoT-based intelligence sharing, where generated rules and attack insights are distributed across multiple nodes to support collaborative and adaptive defense.

8. Novelty of Proposed Solution

- Behavioral intelligence-driven signature generation focusing on attack sequences rather than payload-only pattern matching.
- Integration of unsupervised machine learning with honeypot-derived real-world attack data for adaptive detection.
- Automated conversion of machine learning outputs into deployable IDS and firewall security rules.
- Early containment strategy that mitigates emerging threats without waiting for official vulnerability patches.

9. System Novelty

- **Adaptive Zero-Day Detection:** Learns attack patterns from live honeypot data instead of relying on predefined signatures.
- **Behavior-Aware Analysis:** Combines payload inspection with attacker behavior and attack sequence analysis.
- **Automated Rule Generation:** Automatically generates detection and prevention rules during early stages of an attack.
- **Early Containment Approach:** Detects and mitigates zero-day attacks without patching underlying vulnerabilities.
- **End-to-End Security Platform:** Integrates honeypots, machine learning, rule deployment, and monitoring in a single system.

10. Comprehensive Architecture

Module Mapping

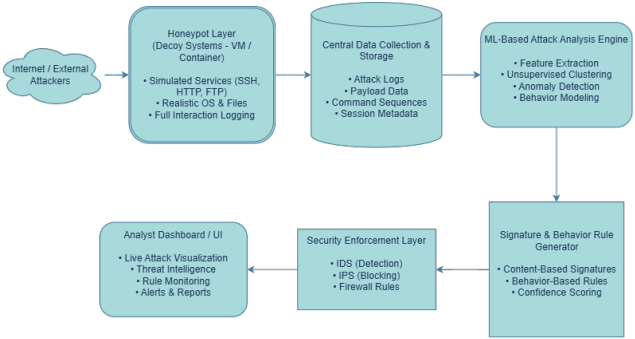
- Honeypot and Data Ingestion
- Analysis and Core Logic
- Dashboard and Integration

DB Schema

- Attack_Session
- Payload_Data
- Attack_Cluster
- Generated_Rules

Key APIs

- `POST /api/ingest`
- `POST /api/analyze`
- `POST /api/rules`



11. Timeline of Workplan

Phase 1: Honeypot Deployment and Structured Data Capture

- Deploy and configure honeypot environment (e.g., SSH/Telnet simulation).
- Enable structured logging of attack sessions, commands, and payload data.
- Validate data collection through controlled attack simulations.

Phase 2: ML-Based Clustering and Rule Generation

- Extract behavioral features from attack sessions to identify patterns and sequences
- Apply clustering and anomaly detection to identify known and zero-day attacks
- Generate behavioral signatures and IDS rules with confidence scoring

Phase 3: Dashboard Development and Integration Testing

- Develop dashboard for real-time visualization of attacks and generated intelligence
- Implement Social IoT-based rule sharing across nodes for collaborative defense.
- Perform attack replay and evaluate system using detection and performance metrics.

12. Implementation - Phase 1

- Deployed Cowrie SSH/Telnet honeypot in a controlled environment.
- Configured honeypot to simulate a real SSH service on port 2222.
- Verified service visibility using Nmap port scanning.
- Performed manual SSH login attempts to simulate attacker behavior.
- Executed automated brute-force attacks using Hydra.
- Captured attacker interactions including login attempts, commands, and IP addresses.
- Stored logs in structured formats (login logs, command logs, IP tracking).
- Generated real-world attack data for further analysis.

Screenshots

```
vrithika@VRITHIKA:~$ nmap -Pn -sV -p 2222 172.20.10.2
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-09 23:23 IST
Nmap scan report for 172.20.10.2
Host is up (0.048s latency).

PORT     STATE SERVICE VERSION
2222/tcp open  ssh     OpenSSH 9.2p1 Debian 2+deb12u3 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.10 seconds
```

Figure: Nmap Scan Result

```
^Cvrithika@VRITHIKA:~$ hydra -l root -P passwords.txt ssh://172.20.10.2:2222 -t 4
Hydra v9.2 (c) 2021 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes
(this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-09 23:28:16
[DATA] max 4 tasks per 1 server, overall 4 tasks, 6 login tries (l:1/p:6), ~2 tries per task
[DATA] attacking ssh://172.20.10.2:2222/
[2222][ssh] host: 172.20.10.2   login: root   password: password
[2222][ssh] host: 172.20.10.2   login: root   password: admin
1 of 1 target successfully completed, 2 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-09 23:28:19
```

Figure: Hydra Brute Force Attack Output

Screenshots

```
vrithika@VRITHIKA:~$ ssh -p 2222 admin@172.20.10.2
admin@172.20.10.2's password:
Permission denied, please try again.
admin@172.20.10.2's password:
Permission denied, please try again.
admin@172.20.10.2's password:
admin@172.20.10.2: Permission denied (publickey,password).
vrithika@VRITHIKA:~$ ssh -p 2222 admin@172.20.10.2
admin@172.20.10.2's password:
Permission denied, please try again.
admin@172.20.10.2's password:
Permission denied, please try again.
admin@172.20.10.2's password:
admin@172.20.10.2: Permission denied (publickey,password).
vrithika@VRITHIKA:~$ ssh -p 2222 user@172.20.10.2
user@172.20.10.2's password:
Permission denied, please try again.
user@172.20.10.2's password:

vrithika@VRITHIKA:~$ ssh -p 2222 ubuntu@172.20.10.2
ubuntu@172.20.10.2's password:
Permission denied, please try again.
ubuntu@172.20.10.2's password:

vrithika@VRITHIKA:~$ ssh -p 2222 test@172.20.10.2
test@172.20.10.2's password:
Permission denied, please try again.
test@172.20.10.2's password:
```

Figure: Manual SSH Login Attempts

Screenshots

```
cowrie@anaswara-HP-Laptop-15s-fq5xxx:~/cowrie$ cat ~/attack_ips.txt
172.20.10.3
172.20.10.4
```

Figure: Captured Attacker IP Addresses

```
cowrie@anaswara-HP-Laptop-15s-fq5xxx:~$ cat ~/attack_logins.txt
2026-02-09T23:09:37.345314Z [HoneyPotSSHTransport,0,172.20.10.4] login attempt [b'root'/b'
amritha123'] succeeded
2026-02-09T23:14:19.841156Z [HoneyPotSSHTransport,1,172.20.10.4] login attempt [b'root'/b'
vrihti'] succeeded
2026-02-09T23:19:58.497434Z [HoneyPotSSHTransport,2,172.20.10.4] login attempt [b'root'/b'
vrithi'] succeeded
2026-02-09T23:23:30.538958Z [HoneyPotSSHTransport,3,172.20.10.3] login attempt [b'root'/b'
1234'] succeeded
2026-02-09T23:27:55.279644Z [HoneyPotSSHTransport,6,172.20.10.3] login attempt [b'root'/b'
1234'] succeeded
2026-02-09T23:28:19.036956Z [HoneyPotSSHTransport,9,172.20.10.4] login attempt [b'root'/b'
123456'] failed
2026-02-09T23:28:19.038432Z [HoneyPotSSHTransport,11,172.20.10.4] login attempt [b'root'/b
'password'] succeeded
2026-02-09T23:28:19.040163Z [HoneyPotSSHTransport,10,172.20.10.4] login attempt [b'root'/b
'admin'] succeeded
2026-02-09T23:28:19.041606Z [HoneyPotSSHTransport,8,172.20.10.4] login attempt [b'root'/b'
root'] failed
2026-02-09T23:30:21.918496Z [HoneyPotSSHTransport,12,172.20.10.3] login attempt [b'root'/b
'1234'] succeeded
2026-02-09T23:31:25.816996Z [HoneyPotSSHTransport,13,172.20.10.3] login attempt [b'root'/b
'1234'] succeeded
```

Figure: Login Attempt Logs

Screenshots

```
cowrie@anaswara-HP-Laptop-15s-fq5xxx:~/cowrie$ cat ~/attack_commands.txt
2026-02-09T23:10:03.144715Z [HoneypotSSHTransport,0,172.20.10.4] CMD: whoami
2026-02-09T23:10:07.444569Z [HoneypotSSHTransport,0,172.20.10.4] CMD: pwd
2026-02-09T23:14:26.494382Z [HoneypotSSHTransport,1,172.20.10.4] CMD: whoami
2026-02-09T23:14:55.063758Z [HoneypotSSHTransport,1,172.20.10.4] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:16:38.490111Z [HoneypotSSHTransport,1,172.20.10.4] CMD: echo -e "password\nadmin\n123456\nroot\nletmein\nwelcome" > passwords.txt
2026-02-09T23:17:07.061083Z [HoneypotSSHTransport,1,172.20.10.4] CMD: hydra -l root -P passwords.txt ssh://172.20.10.2:2222 -t 4
2026-02-09T23:19:17.732541Z [HoneypotSSHTransport,1,172.20.10.4] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:20:12.211745Z [HoneypotSSHTransport,2,172.20.10.4] CMD: whoami
2026-02-09T23:20:44.060030Z [HoneypotSSHTransport,2,172.20.10.4] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:21:08.779539Z [HoneypotSSHTransport,2,172.20.10.4] CMD: exit
2026-02-09T23:52:07.078393Z [HoneypotSSHTransport,3,172.20.10.3] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:28:11.458884Z [HoneypotSSHTransport,6,172.20.10.3] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:28:50.883484Z [HoneypotSSHTransport,6,172.20.10.3] CMD:
2026-02-09T23:28:50.884624Z [HoneypotSSHTransport,6,172.20.10.3] CMD:
2026-02-09T23:28:51.190134Z [HoneypotSSHTransport,6,172.20.10.3] CMD:
2026-02-09T23:28:51.191692Z [HoneypotSSHTransport,6,172.20.10.3] CMD:
2026-02-09T23:28:51.501045Z [HoneypotSSHTransport,6,172.20.10.3] CMD:
2026-02-09T23:28:51.502165Z [HoneypotSSHTransport,6,172.20.10.3] CMD:
2026-02-09T23:28:51.908866Z [HoneypotSSHTransport,6,172.20.10.3] CMD:
2026-02-09T23:29:31.027621Z [HoneypotSSHTransport,6,172.20.10.3] CMD: exit
2026-02-09T23:30:41.374042Z [HoneypotSSHTransport,12,172.20.10.3] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:31:03.804084Z [HoneypotSSHTransport,12,172.20.10.3] CMD: exit
2026-02-09T23:31:34.726986Z [HoneypotSSHTransport,13,172.20.10.3] CMD: apt update
2026-02-09T23:31:34.729138Z [HoneypotSSHTransport,13,172.20.10.3] CMD: apt install nmap -y
2026-02-09T23:32:15.688514Z [HoneypotSSHTransport,13,172.20.10.3] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:32:42.008316Z [HoneypotSSHTransport,13,172.20.10.3] CMD: apt update
2026-02-09T23:32:42.011623Z [HoneypotSSHTransport,13,172.20.10.3] CMD: apt install --reinstall nmap -y
2026-02-09T23:33:16.819546Z [HoneypotSSHTransport,13,172.20.10.3] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:33:41.295313Z [HoneypotSSHTransport,13,172.20.10.3] CMD: apt install --reinstall libpcap0.8 -y
2026-02-09T23:34:45.195373Z [HoneypotSSHTransport,13,172.20.10.3] CMD: nmap -sV -p 2222 172.20.10.2
2026-02-09T23:34:46.937398Z [HoneypotSSHTransport,13,172.20.10.3] CMD: echo -e "password\nadmin\n123456\nroot\nletmein\nwelcome" > passwords.txt
2026-02-09T23:35:07.213174Z [HoneypotSSHTransport,13,172.20.10.3] CMD: hydra -l root -P passwords.txt ssh://172.20.10.2:2222 -t 4
2026-02-09T23:35:31.275917Z [HoneypotSSHTransport,13,172.20.10.3] CMD: apt update
2026-02-09T23:35:31.280847Z [HoneypotSSHTransport,13,172.20.10.3] CMD: apt install hydra -y
2026-02-09T23:35:51.859406Z [HoneypotSSHTransport,13,172.20.10.3] CMD: hydra -l root -P passwords.txt ssh://172.20.10.2:2222 -t 4
2026-02-09T23:35:55.853259Z [HoneypotSSHTransport,13,172.20.10.3] CMD: exit
```

Figure: Command Execution Logs

Screenshots

```
cowrie@anaswara-HP-Laptop-15s-fq5xxx:~/cowrie$ # Create a detailed report
echo "=== HYDRA BRUTE FORCE ATTACK - IP 172.20.10.5 ===" > ~/attack_172.20.10.5_analysis.tx
t
echo "" >> ~/attack_172.20.10.5_analysis.txt
echo "Passwords Attempted:" >> ~/attack_172.20.10.5_analysis.txt
grep "172.20.10.5" var/log/cowrie/cowrie.log | grep "login attempt" >> ~/attack_172.20.10.5
_analysis.txt

# View it
cat ~/attack_172.20.10.5_analysis.txt

# Copy to Desktop
exit
cp /home/cowrie/attack_172.20.10.5_analysis.txt ~/Desktop/
=== HYDRA BRUTE FORCE ATTACK - IP 172.20.10.5 ===

Passwords Attempted:
2026-02-10T00:28:44.655674Z [HoneyPotSSHTransport,23,172.20.10.5] login attempt [b'root'/b'
hello'] succeeded
2026-02-10T00:33:58.304934Z [HoneyPotSSHTransport,24,172.20.10.5] login attempt [b'root'/b'
hi'] succeeded
2026-02-10T00:45:34.407659Z [HoneyPotSSHTransport,29,172.20.10.5] login attempt [b'root'/b'
admin'] succeeded
2026-02-10T00:45:34.409192Z [HoneyPotSSHTransport,30,172.20.10.5] login attempt [b'root'/b'
root'] failed
2026-02-10T00:45:34.409987Z [HoneyPotSSHTransport,27,172.20.10.5] login attempt [b'root'/b'
password'] succeeded
2026-02-10T00:45:34.411114Z [HoneyPotSSHTransport,28,172.20.10.5] login attempt [b'root'/b'
123456'] failed
2026-02-10T00:54:32.509988Z [HoneyPotSSHTransport,33,172.20.10.5] login attempt [b'root'/b'
abc123'] succeeded
2026-02-10T00:54:32.528409Z [HoneyPotSSHTransport,34,172.20.10.5] login attempt [b'root'/b'
welcome'] succeeded
2026-02-10T00:54:32.532091Z [HoneyPotSSHTransport,32,172.20.10.5] login attempt [b'root'/b'
qwerty'] succeeded
2026-02-10T00:54:32.534010Z [HoneyPotSSHTransport,35,172.20.10.5] login attempt [b'root'/b'
letmein'] succeeded
logout
```

Figure: Attack Analysis Report Generation

Screenshots

```
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-09 19:15:35
amrit@AMRITHA: $ cat > apple.txt << EOF
letmein
welcome
qwerty
abc123
password123
admin123
EOF
amrit@AMRITHA: $ hydra -l root -P apple.txt ssh://172.20.10.2:2222 -t 4 -vV
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or fo
gal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-09 19:24:32
[DATA] max 4 tasks per 1 server, overall 4 tasks, 6 login tries (l:1/p:6), ~2 tries per task
[DATA] attacking ssh://172.20.10.2:2222/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://root@172.20.10.2:2222
[INFO] Successful, password authentication is supported by ssh://172.20.10.2:2222
[ATTEMPT] target 172.20.10.2 - login "root" - pass "letmein" - 1 of 6 [child 0] (0/0)
[ATTEMPT] target 172.20.10.2 - login "root" - pass "welcome" - 2 of 6 [child 1] (0/0)
[ATTEMPT] target 172.20.10.2 - login "root" - pass "qwerty" - 3 of 6 [child 2] (0/0)
[ATTEMPT] target 172.20.10.2 - login "root" - pass "abc123" - 4 of 6 [child 3] (0/0)
[2222][ssh] host: 172.20.10.2   login: root   password: abc123
[2222][ssh] host: 172.20.10.2   login: root   password: welcome
[2222][ssh] host: 172.20.10.2   login: root   password: qwerty
[STATUS] attack finished for 172.20.10.2 (waiting for children to complete tests)
[2222][ssh] host: 172.20.10.2   login: root   password: letmein
1 of 1 target successfully completed, 4 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-09 19:24:33
amrit@AMRITHA: $
```

Figure: Custom Password List and Hydra Attack

Screenshots


```
[
  {
    "ip": "172.20.10.5",
    "failed_attempts": 4,
    "successful_attempts": 4,
    "total_attempts": 8,
    "duration": "few seconds",
    "attack_type": "brute force pattern"
  },
  {
    "ip": "172.20.10.3",
    "failed_attempts": 0,
    "successful_attempts": 1,
    "total_attempts": 1,
    "duration": "instant",
    "attack_type": "single login attempt"
  }
]
```


Figure: Reconstructed Session Data

References I

1. R. R. Patel and C. S. Thaker, “Zero-day attack signatures detection using honeypot,” *Journal of Computer Science (IJCSIS)*, Sept. 2017. https://www.academia.edu/35814683
2. H. Hindy, R. Atkinson, C. Tachtatzis, J.-N. Colin, E. Bayne, and X. Bellekens, “Utilising deep learning techniques for effective zero-day attack detection,” *Electronics*, vol. 9, no. 10, pp. 1–24, 2020. https://doi.org/10.3390/electronics9101684
3. Y. Guo, “A review of machine learning-based zero-day attack detection: Challenges and future directions,” *Computer Communications*, vol. 198, pp. 175–185, 2023. https://doi.org/10.1016/j.comcom.2022.11.001
4. M. Balamurugan, “AI-enhanced honeypots for zero-day exploit detection and mitigation,” *International Journal for Multidisciplinary Research*, vol. 6, no. 6, pp. 1–8, Nov.–Dec. 2024. https://doi.org/10.36948/ijfmr.2024.v06i06.32866
5. A. Blaise, M. Bouet, and V. Conan, “Detection of zero-day attacks: An unsupervised port-based approach,” *Computer Networks*, vol. 180, pp. 1–15, 2020. https://doi.org/10.1016/j.comnet.2020.107391

References II

6. Y. Yu, J. Li, and J. Li, “An adaptable deep learning-based intrusion detection system to zero-day attacks,” *Computer Networks*, vol. 196, pp. 1–12, 2021. https://doi.org/10.1016/j.neucom.2024.128073

Thank You